The National Institute of Justice Recidivism Challenge is a multi-stage project comprised of five assignments, each aligned with a stage of the Data Science Analysis Process. Assignments 1 and 4 are individual tasks, while Assignments 2, 3, and 5 are team-based.

The Data Science Analytics Process

1. Understand
2. The Data
3. Model
4. Understand
5. Communicate

Assignment 1: Understand

The individual assignment requires students to analyze and understand their data source.

Student Objectives:

Research and understand the premise of the project
Research and understand key/domain vocabulary
Understand the research question
Ask in-depth questions about the data
Perform basic Python functions to gain a better understanding of the data
Use libraries such as Matplotlib and Seaborn to create basic visualizations

Assignment 2: The Data

In this group project, students will build on their understanding from Assignment 1. Working in teams of 3-4, they will clean and prepare their data for future assignments and perform an exploratory analysis to identify important areas within their data.

Student Objectives:

Find areas where data is "dirty" and determine ways to clean it
Use Python to clean and prepare data
Use Python to analyze data
Use libraries such as Matplotlib and Seaborn to create visualizations
Build teamwork and communication skills

Assignment 3: Model

In the same groups, students will use their knowledge from the previous assignment to create three machine-learning models of their choice.

Student Objectives:

Demonstrate a clear understanding of machine learning models
Use Python to create machine learning models
Use Python to create confusion matrices
Build teamwork and communication skills

Assignment 4: Understand

Assignment 4 is an individual assignment where students will write a short paper detailing their team's models from Assignment 3. This paper should

briefly explain how each model works, provide an explanation of the confusion matrix, compare the three models, and determine which model is the best fit. Remind students that the audience of their paper is members of the NIJ, not other data scientists.

Student Objectives:

- Demonstrate a clear understanding of machine learning models
- Demonstrate a clear understanding of the confusion matrices
- Express critical thinking skills in the comparison and selection of models
- Convert technical language into layman's terms
- Build written communication skills

Peer Evals:
Each student should complete a Peer Evaluation and Statement of Contribution for every group member, to be submitted before the team presents Assignment 5. These evaluations should be taken into consideration when grading.

Assignment 5: Communicate

For the final assignment, students will return to their original teams and create a presentation about their project using the "Answer First" format. Presentations will be given to the class as if they were presented to members of the NIJ.

Student Objectives:

- Demonstrate a full understanding of the project and its assignments
- Improve public speaking/verbal communication skills
- Build teamwork and communication skills